Sumaiya Dharsi

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EDUCATION

Northeastern University, College of Engineering, Boston, MA

- Candidate for Bachelor of Science in Mechanical Engineering and Design, Anticipated May 2028
- GPA: 3.3/4.00
- Associations: Theme Park Engineering Club, Society of Women Engineers, Undergraduate South Asian Organization, Islamic Society (ISNU)
- Relevant Coursework: Cornerstone of Engineering, Statics, Thermodynamics, Mechanics of Materials, Dynamics, Designing Interactive Experiences

SKILLS

- Programming Languages: C++, HTML, CSS, MATLAB
- CAD & Mechanics: SolidWorks, Arduino, 3D printing (SLA/FDM), Laser Cutting, Soldering
- Digital Tools: Microsoft Suite, Adobe Suite
- Spoken Languages: English (native), Chinese (conversational), Gujarati (elementary)

EXPERIENCE

10 Beauty, Burlington, MA

Mechanical Engineer Co-op, January to June 2026

- Prototyped components for a robotic manicure machine using SolidWorks, collaborating with the engineering team to enhance consistency and reliability across design iterations.
- Performed force and viscosity testing on material samples, analyzing results to drive iterative design refinements for the next build.
- Presented testing results and design recommendations to the engineering team to guide prototype development priorities.

Digitech Printers, East Meadow, NY

Design Intern, Summer 2023

- Produced custom apparel designs for 7+ clients using heat press and vinyl application techniques.
- Gained hands-on exposure to design workflow and production processes in a professional print environment.

PROJECTS

MatchaBot

Personal Project, Spring 2026 to Present

- Engineered an automated matcha whisking device that replicates the traditional chasen's zigzag motion, performing gear calculations and mechanical design to convert rotational motion into whisking action.
- Integrated 3D-printed mechanisms with an Arduino microcontroller to control motion and achieve smooth, froth-free matcha.

***Flexir*: Hand Physical Therapy Tool**

Cornerstone of Engineering Project, Spring 2025

- Designed and prototyped a wearable rehabilitation device in SolidWorks with 3D-printed components, integrating Arduino sensors to track hand mobility and share progress data through MATLAB.
- Conducted and analyzed 10 hand-movement tests in MATLAB to validate sensor accuracy, refining functionality for at-home therapy use.

***Wreck-It Ralph*: Building Disaster**

Cornerstone of Engineering Project, Fall 2024

- Developed an interactive educational game using C++, Arduino, and laser-cut components to introduce children to structural engineering concepts.
- Conducted user experience testing, gathering feedback to refine gameplay and improve learning outcomes.
- Gained practical prototyping and teamwork experience through collaborative, hands-on design.

INTERESTS

- Painting, Volleyball, Soccer, Pottery